

Constraints

Marketing Agent POC

Technical Constraints

- Budget: Zero. No API spend, no paid subscriptions. Every component must have a free tier or run locally.
 - Hardware: Developer laptop (8GB RAM, no dedicated GPU). Local model inference must run on CPU.
 - Vision inference: Local vision models (LLaVA, Qwen-VL) require 8-16GB VRAM and run at <3 tokens/s on CPU -- too slow. Cloud vision required for image analysis.
 - Model hosting: Ollama only for local text generation. No Docker, no self-hosted inference server.
 - Dependencies: pip install only. No external services beyond Ollama and an OpenRouter account.
-

Functional Constraints

- Input: Product image (PNG/JPG/WEBP, ?16MB) + short description (15-40 words). No other required input.
 - Output scope: Audience persona, two copy variants, compliance flags, CTA scores. No landing page, no theme selection, no evaluation scoring in the POC.
 - Character limit: Each ad copy variant ?150 characters primary text.
 - Compliance: Must flag guarantee claims, health claims, superlatives, urgency language, scientific claims, weight loss claims. No false negatives acceptable on these categories.
 - Pipeline time: Complete campaign generation in under 120 seconds.
-

Design Constraints

- No session state: The refinement loop must work without server-side sessions -- the full output state travels with each /refine request.
 - No frontend framework: Plain HTML/CSS/JS only. No React, no build step.
 - Guardrails in code: Compliance logic cannot live in a prompt. A prompt-based compliance check can hallucinate. Rules must be deterministic.
 - Visible routing: Agent routing decisions must be shown in the UI. The assignment explicitly requires visible agent collaboration.
-

Scope Constraints

- POC only: The system demonstrates the architecture, not a production deployment. No authentication, no multi-tenancy, no rate limiting.
 - Single product per run: One image, one description, one campaign. Batch generation is out of scope.
 - Deferred: Theme selection (FR3), landing page generation (FR6), evaluation scoring (FR10).
-

Educational Constraints

- Decision logs required: Every significant architectural choice must be documented with a research question, evidence, and what it unlocks.
- Ethics required: The assignment brief explicitly states "your decision log must grapple with this reality" (job displacement). This is a graded requirement.
- LO coverage required: Evidence must exist across all five LO stages -- Analyzing, Advising, Designing, Realizing, Managing.